

When several tones become one

Background

Consider $x(t) = \sum A[i] \cos(2\pi f[i] t)$, with zero phase. Under sampling, each cosine folds to an equivalent frequency in $[0, f_s/2]$. Components landing at the same frequency add their signed amplitudes. Opposite amplitudes can cancel. A zero-frequency component is a constant (DC).

Your task

Return the folded multitone model after merging collisions and removing groups whose total amplitude is exactly zero. Then evaluate the original and folded models at the same sample instants and report their maximum difference. The provided plot displays both continuous models and their samples.

Function contract and assumptions

`collapse_tones(freqs, amps, fs)` returns `(alias_freqs, alias_amps)`, sorted by ascending frequency. Use integer frequencies, integer signed amplitudes, and positive even integer `fs`; duplicate frequencies are allowed. If all groups cancel, return two empty arrays. `evaluate_tones(t, freqs, amps)` returns the signal values at `t`; an empty model returns zeros. A folding helper is provided.

Example / correctness check

`freqs=[3,17,23,7,13,20]`, `amps=[2,1,-1,3,-3,4]`, `fs=20` gives `alias_freqs=[0,3]`, `alias_amps=[4,2]`. The 7 Hz group cancels; the 3 Hz group sums to 2. With `N=40`, sample error should be below $1e-10$.

Exam instructions

Complete only the TODO functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF pages 5 and 22; collision addition follows linearity and is explained above.